



Answer four out of five questions

- $$-(1, 2) \cup (2, 3)$$

ii. Supremum of the set A .

- $$A = \{1 - 1/n \mid n \in \mathbb{N}\}$$

- $$\sup(f + g)[D] \leq \sup f[D] + \sup g[D].$$

- $x < n$
 $1/n$
 $x < 1/n$
 $x > 0$
 $x < 1/n$
 $x > 0$
 $x < 1/n$
 $x > 0$

$$x' \in x \subset 1$$

- circles

- ii. Let (x_n) be a monotonically increasing sequence. Show that (x_n) is convergent if and only if x_n is bounded.

- $9 \frac{1}{2} \times 1 \frac{1}{2} \times 1 \frac{1}{2}$

$$x_n = 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n}$$

is not convergent. State carefully any theorems that you may use.

3. (a) State and prove the sequential criterion for the existence of a limit.

(b) Let $f : D \rightarrow \mathbb{R}$ and $g : D \rightarrow \mathbb{R}$ be functions and let a be a limit point of D . Show that

$$\lim_{x \rightarrow a} (fg)(x) = \left(\lim_{x \rightarrow a} f(x) \right) \left(\lim_{x \rightarrow a} g(x) \right).$$

(c) Prove that

$$\lim_{x \rightarrow 0} \frac{2x^2 - 3x + 1}{x - 1} = -1,$$

using the $\varepsilon - \delta$ definition.

(d) Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined by

$$f(x) = \begin{cases} \sin(1/x) & \text{if } x \neq 0, \\ 0 & \text{if } x = 0. \end{cases}$$

Show that the limit of f at 0 does not exist.

4. (a) Define what it means for a function $f : D \rightarrow \mathbb{R}$ to be *continuous* at a point $x \in D$.

(b) State the Extreme Value Theorem.

(c) i. Let $f : [a, b] \rightarrow \mathbb{R}$ be a continuous function and suppose that $f(a) < 0 < f(b)$. Show that there exists a point $c \in (a, b)$ such that $f(c) = 0$.

(Hint: Set $S = \{x \in [a, b] \mid f(x) \leq 0\}$ and $c = \sup S$, and show that $f(c) = 0$.)

ii. Hence prove that the equation $5^x = x^4$ has at least one real root.

(d) Show that the function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$f(x) = \begin{cases} x & \text{if } x \in \mathbb{Q}, \\ 0 & \text{otherwise,} \end{cases}$$

is continuous only at 0.

5. (a) State and prove the Rolle's theorem.

(b) Hence deduce the following result: Let $f : [a, b] \rightarrow \mathbb{R}$ be a function that is continuous on $[a, b]$ and differentiable on (a, b) , then there exists a point $c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

(c) Derive the following inequality

$$\frac{x-1}{x} < \ln(x) < x-1,$$

provided that $x > 1$.

(d) Find the following limits. You must clearly cite any theorem you may use.

i. $\lim_{x \rightarrow 0} \left(\frac{1}{x} - \frac{1}{\sin x} \right).$

ii. $\lim_{x \rightarrow 0^+} (1 + 2x)^{1/x}.$